

Scalability & Future Plan

Date	29/09/2026
Team ID	UNM1425LegalEase: AI-Powered Legal Document Generator
Project Name	LegalEase: AI-Powered Legal Document Generator
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Single-instance deployment	Scaling concurrent users may require additional backend instances	High
2	External AI/API dependency	Availability and rate limits can affect generation	High
3	File-based document processing	Large concurrent exports can increase resource usage	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	FastAPI backend + Streamlit frontend; demo uses a connected service	Add load balancing, multiple API instances and autoscaling.
2	Data Storage	Generated documents are produced for export during requests	Use managed object storage and persistent storage for larger usage.
3	Performance	AI generation and document formatting run per request	Add caching, async/background jobs and performance monitoring.
4	Security	API key uses environment configuration; sensitive-data handling is planned	Add secrets management, access control, encryption and audit logging.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Validate current demo workflow and document generation	Current project cycle	Working baseline for further scaling

Phase 2	Add authentication, user accounts and secure document history	Next development cycle	Personalized access and document management
Phase 3	Scale backend with managed storage, caching and monitoring	Later development cycle	Higher concurrency and reliability
Phase 4	Add multilingual support, richer legal templates and analytics	Future enhancement cycle	Broader accessibility and feature coverage